Software Requirmnets Specification

For

Tour -Visit Planner (DayTripGO)

A Web-Based Visit Planning System for Wadduwa and Surrounding Areas

Document Version: 2.0

Student Name - De Silva P.K.M
Registration Number - E2410884
Date: 05.09.2026

Revision History

Version	Date	Author	Description of Change
1.0	21.08.2026	P.K.M.DE Silva	Initial draft of SRS based on approved project proposal.
2.0	05.09.2026	P.K.M.DE Silva	Final draft of SRS based on Initial and peer review

Table of Contents

Revision History 2
Table of Contents 3
1.1 Purpose 5
1.2 Document Conventions 5
1.3 Intended Audience and Reading Suggestions 5
1.4 Project Scope 5
1.5 References 6
2. Overall Description 6
2.1 Product Perspective 6
2.2 Product Functions 6
2.3 User Classes and Characteristics 7
2.4 Operating Environment 7
2.5 Design and Implementation Constraints 7
2.6 User Documentation 8
2.7 Assumptions and Dependencies 8
3. External Interface Requirements 8
3.1 User Interfaces 8
3.2 Hardware Interfaces 9
3.3 Software Interfaces 9
3.4 Communications Interfaces 9
4. System Features (Functional Requirements) 10
4.1 Place Listing and Category Filtering 10
4.2 Place Detail View 10
4.3 Map View 11
4.4 One-Day Visit Plan Builder 11
4.5 Administrator Authentication 11
4.6 Administrator Place Management 12
5. Non-Functional Requirements 12
5.1 Performance Requirements 12
5.2 Safety Requirements 12
5.3 Security Requirements 12
5.4 Software Quality Attributes 13
5.1 Safety Requirements 13
5.2 Security Requirements 13
5.3 Software Quality Attributes 14

6. Data Requirements 14
6.1 Place Entity 15
6.2 Category Entity 15
6.3 VisitPlan and VisitPlanItem Entities 15
6.4 Administrator Entity 15
7. Requirements Traceability 16
8. Appendices 17
Appendix A: Glossary 17
Appendix B: Open Issues 17
B.1 Use Case Diagram 17
Figure 1: Use Case Diagram for DayTripGO 17
B.3 Class Diagram 18
Figure 2: Class Diagram for DayTripGO 18
B.3 Sequence Diagram 19
Figure 3: Sequence Diagram — Build a One-Day Visit Plan 19
B.4 Entity-Relationship Diagram 19
Figure 4: Entity-Relationship Diagram for DayTripGO 20
Appendix C: Open Issues 21

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Local Tourist Day-Visit Planner and Information System, a web-based application that helps tourists discover places of interest within 25 km of Wadduwa, Sri Lanka, and build a simple one day visit plan. This document is intended for the developer, the module supervisor/assessor, and the four consulted stakeholders, and it forms the basis for design, implementation and testing of the system under module ITE2953.

1.2 Document Conventions

Requirements are labelled using the prefixes FR (Functional Requirement) and NFR (Non-Functional Requirement), each followed by a sequential number, e.g. FR-01. The keywords "shall" and "must" indicate mandatory requirements; "should" indicates a desirable but non-mandatory requirement. Priority is expressed as High, Medium or Low, where High requirements are necessary for the system to fulfil its core purpose.

1.3 Intended Audience and Reading Suggestions

- Developer: primary audience; uses Sections 2–6 for design and implementation guidance.
- Module Supervisor / Assessor: uses Sections 1–3 to verify the requirements match the approved proposal, and Section 4 to trace testable requirements.
- Stakeholders (tourist representatives): may review Section 2 (Overall Description) and Section 4 (Functional Requirements) in plain language to confirm their needs have been captured correctly.

1.4 Project Scope

The system scope, as approved in the project proposal, covers: browsing and filtering at least 10 verified places of interest within 25 km of Wadduwa; viewing place details including description, opening times, travel tips and distance; viewing each place on a map; building and viewing a simple one-day visit plan; and an administrator interface for adding, editing and deleting place records. Booking, payment, multi-day itineraries and transport/accommodation reservations are explicitly out of scope.

1.5 References

[1] Project Proposal: "Local Tourist Day-Visit Planner and Information System," ITE2953 Programming Group Project 25S1.

[2] IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications (used as the structural template for this document).

2. Overall Description

2.1 Product Perspective

The Tourist Day-Visit Planner is a new, standalone web application; it is not a component of, or replacement for, an existing system. It consists of a public tourist-facing interface, a restricted administrator interface, and a shared relational database of places, categories and visit plans. The system depends on a third-party map-embedding service (e.g. Google Maps or OpenStreetMap) to plot place locations, and this is treated as an external interface rather than something built from scratch (see Section 5).

2.2 Product Functions

At a high level, the system provides the following functions, summarised here and detailed as individual requirements in Section 4:

- List and filter places of interest by category (Tourist).
- View detailed information for a single place, including a map of its location (Tourist).
- Add places to, remove places from, and view a one-day visit plan (Tourist).
- Authenticate as an administrator (Administrator).
- Create, update and delete place records (Administrator).

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Tourist	Domestic or foreign visitor staying in or near Wadduwa who browses places of interest and builds a one-day visit plan. Primary, unauthenticated user.	Low to moderate; assumed familiar with basic web browsing on desktop or mobile.
Administrator	Person responsible for keeping place-of-interest data accurate; logs in through a restricted interface to add, edit or delete records.	Moderate; comfortable with web forms, no coding knowledge required.

2.4 Operating Environment

- Client: modern desktop and mobile web browsers (Chrome, Firefox, Edge, Safari, latest two major versions).
- Server: Node.js runtime with an Express-based backend (or the module-approved equivalent).
- Database: PostgreSQL relational database hosted on superbase.
- Hosting: development/demo environment suitable for local or low-cost cloud hosting for the duration of the module; production-grade hosting is out of scope.

2.5 Design and Implementation Constraints

- The technology stack must match the stack approved for ITE2953 (see Section 1.4 of the proposal); any deviation requires supervisor approval.
- The system must be delivered without payment, booking or reservation functionality, per the approved scope.
- The place dataset must contain a minimum of 10 verified places within25 km of Wadduwa at the point of submission.

- Development is carried out by a single student within the timeframe of one academic module, which constrains the depth of testing and the size of the feature set.

2.6 User Documentation

A short user guide (or in-app help text) covering how to filter places, view details, build a one-day plan, and how an administrator logs in and manages records will be delivered alongside the system, in line with standard ITE2953 submission requirements.

2.7 Assumptions and Dependencies

- It is assumed that opening times and travel tips supplied by place owners/authorities do not change frequently enough to require real-time synchronisation.
- The system depends on an external map-embedding API remaining available and free-to-use at the tier required for a student project.
- It is assumed that the four consulted stakeholders are broadly representative of the primary Tourist user class for the purpose of requirements validation.

3. External Interface Requirements

3.1 User Interfaces

- Home, Place Listing page - displays places of interest as a filterable list or grid, with category filter controls.
- Place Detail page - displays a single place's description, opening times, travel tips, distance from Wadduwa, and an embedded map.
- Visit Plan page - displays the tourist's currently selected places for a one-day plan, in a single consolidated view.
- Administrator Login page - a restricted form for administrator authentication.
- Administrator Dashboard - a form based interface for adding, editing and deleting place records.

Detailed wireframes/mockups are to be produced in Figma during the design phase and referenced here as figures once available (see List of Figures).

3.2 Hardware Interfaces

No dedicated hardware is required. The system relies on the client device's existing display, input and network hardware; no sensors, printers or other peripherals are used.

3.3 Software Interfaces

Interface	Type	Purpose
Map Service API	Third-party web API	Renders an embedded map and marker for each place's location on the Place Detail page.
PostgreSQL Database	Relational database	Stores place, category, visit-plan and administrator account data; accessed by the backend only.
Backend REST API	Internal API	Exposes endpoints consumed by the frontend for listing, filtering, detail retrieval, visit-plan management and administrator CRUD operations.

3.4 Communications Interfaces

All communication between the client browser and the backend occurs over HTTPS using standard HTTP request/response semantics (REST). No email, messaging, or real-time push protocols are required by the current scope.

4. System Features (Functional Requirements)

This section describes each system feature and its associated functional requirements, grouped by feature area. Priority follows the MoSCoW convention: Essential (Must Have), Desirable (Should Have), Optional (Could Have).

4.1 Place Listing and Category Filtering

Description: Allows a tourist to browse all places of interest and narrow the list down by category (religious, nature, heritage, cultural).

ID	Requirement Description	Priority
FR-1	The system shall display a list of all places of interest on the home page, showing at minimum a photo, name, category and distance from Wadduwa for each place.	Essential
FR-2	The system shall allow the tourist to filter the displayed places by one category at a time.	Essential
FR-3	The system shall allow the tourist to clear an active filter and return to viewing all places.	Desirable
FR-4	The system shall indicate, where available, whether a place is currently open based on its stored opening hours.	Optional

4.2 Place Detail View

Description: Allows a tourist to view full details of a selected place before deciding to add it to a visit plan.

ID	Requirement Description	Priority
FR-5	The system shall display, for a selected place, its description, opening hours, travel tips and distance from Wadduwa.	Essential
FR-6	The system shall provide a control that allows the tourist to add the selected place directly to their current one-day visit plan.	Essential
FR-7	The system shall display a representative photo of the place on the detail page, where available.	Desirable

4.3 Map View

Description: Allows a tourist to see where a place is located relative to Wadduwa on an interactive map.

ID	Requirement Description	Priority
FR-8	The system shall display the geographic location of a selected place as a marker on an embedded map, using its stored latitude and longitude.	Essential
FR-9	The system shall centre and appropriately zoom the map view on the selected place's coordinates.	Desirable

4.4 One-Day Visit Plan Builder

Description: Allows a tourist to collect selected places into a single consolidated plan for a day visit.

ID	Requirement Description	Priority
FR-10	The system shall allow a tourist to add multiple places to a single one-day visit plan.	Essential
FR-11	The system shall allow a tourist to remove a previously added place from the visit plan.	Essential
FR-12	The system shall display all places currently in the visit plan in one consolidated view.	Essential
FR-13	The system shall allow a tourist to reorder the places within their visit plan.	Desirable

4.5 Administrator Authentication

Description: Restricts place-management functionality to authorised administrators only.

ID	Requirement Description	Priority
FR-14	The system shall require an administrator to log in with a valid username and password before accessing any place-management function.	Essential
FR-15	The system shall deny access to administrator functions for any unauthenticated request.	Essential

4.6 Administrator Place Management

Description: Allows an authenticated administrator to keep the catalogue of places accurate and current.

ID	Requirement Description	Priority
FR-16	The system shall allow an authenticated administrator to add a new place record, including name, category, description, opening hours, travel tips, distance, coordinates and an image.	Essential
FR-17	The system shall allow an authenticated administrator to edit an existing place record.	Essential
FR-18	The system shall allow an authenticated administrator to delete a place record.	Essential
FR-19	The system shall validate that all required fields are provided before a place record is saved, and shall display an error message if validation fails.	Desirable

5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement Description	Priority
NFR-1	The system shall load the home page place listing within 3 seconds under normal conditions on a standard 4G mobile connection.	Essential
NFR-2	The system shall support at least 50 concurrent users without a noticeable degradation in response time, sufficient for classroom demonstration and evaluation.	Desirable
NFR-3	Filtering places by category shall update the displayed list within 1 second of the tourist's selection.	Desirable

5.2 Safety Requirements

ID	Requirement Description	Priority
NFR-4	The system shall not provide any functionality that could mislead a tourist into believing a booking, payment or reservation has been made, since these features are explicitly out of scope.	Essential

5.3 Security Requirements

ID	Requirement Description	Priority
NFR-5	Administrator passwords shall be stored using a secure hashing algorithm and never stored or transmitted in plain text.	Essential

NFR-6	All administrator management endpoints shall verify a valid authentication session before executing any add, edit or delete operation.	Essential
NFR-7	The system shall not collect or store any personally identifiable information from tourist users, since visit plans are session-based and do not require an account.	Desirable

5.4 Software Quality Attributes

ID	Requirement Description	Priority
NFR-8	Usability: A first-time tourist user shall be able to view a place's details and add it to a visit plan within 3 taps/clicks from the home page.	Essential
NFR-9	Reliability: The system shall handle a failed data request (e.g., network loss) by displaying a clear error message rather than crashing or showing a blank page.	Desirable
NFR-10	Maintainability: The codebase shall be organised into modular, reusable components (frontend) and services (backend) to simplify future feature additions.	Desirable
NFR-11	Portability: The system shall be accessible through any modern browser (Chrome, Firefox, Safari, Edge) without requiring installation of additional software.	Essential
NFR-12	Scalability: The underlying database schema shall support additional categories or an expanded geographic radius without requiring structural redesign.	Optional

5.1 Safety Requirements

The system handles no payment, medical, or safety-critical data; no dedicated safety requirements beyond standard data-handling practice apply.

5.2 Security Requirements

- Only authenticated administrators shall be able to create, edit or delete place records (see NFR-04).
- All administrator authentication traffic shall be transmitted over HTTPS.

- Tourist-facing browsing shall not require account creation or collect personal data beyond what is strictly needed for the visit-plan feature (which is session-based, not account-based).

5.3 Software Quality Attributes

- Reliability: core browsing and filtering functions (FR-01–FR-05) shall function correctly for all records in the initial dataset.
- Portability: the backend shall run on any standard Node.js hosting environment without code changes.
- Testability: every functional requirement in Section 4 shall map to at least one test case in the system test plan.

6. Data Requirements

The core entities are Place, Category, VisitPlan, VisitPlanItem and Administrator. The Place table, the entity most central to the system, is detailed below as an example; the remaining entities follow the same convention and will be finalised as an entity-relationship diagram during design (see List of Figures).

6.1 Place Entity

Field	Type / Constraint	Description
place_id	INT, PK, auto-increment	Unique identifier for a place.
name	VARCHAR(150), NOT NULL	Name of the place of interest.
category_id	INT, FK -> category.category_id	Links the place to a category.
description	TEXT, NOT NULL	Descriptive text shown on the detail page.
opening_hours	VARCHAR(100)	Opening times, free text (e.g. "6:00 AM – 8:00 PM daily").
travel_tips	TEXT	Practical tips for visiting.
distance_km	DECIMAL(4,1), NOT NULL	Approximate distance from Wadduwa in kilometres.
latitude	DECIMAL(9,6), NOT NULL	Latitude used for map display.
longitude	DECIMAL(9,6), NOT NULL	Longitude used for map display.
created_at	DATETIME	Record creation timestamp.
updated_at	DATETIME	Record last-updated timestamp.

6.2 Category Entity

- category_id (INT, PK, auto-increment)
- category_name (VARCHAR(50), NOT NULL, UNIQUE) — e.g. Nature, Religious, Heritage, Cultural

6.3 VisitPlan and VisitPlanItem Entities

- VisitPlan: plan_id (PK), session_id or device token, created_at.
- VisitPlanItem: item_id (PK), plan_id (FK), place_id (FK), sort_order — links selected places to a plan in a chosen order.

6.4 Administrator Entity

- admin_id (PK), username (UNIQUE, NOT NULL), password_hash (NOT NULL), created_at

7. Requirements Traceability

This table maps each project objective from the approved proposal to the requirement(s) in this SRS that realise it, to support verification that the SRS is complete relative to the proposal.

Project Objective	Related Requirement(s)
Identify ≥10 verified places within 25 km of Wadduwa	FR-01, FR-04, Section 6.1
Elicit and validate requirements through stakeholder consultation	Section 2.3, Appendix A (proposal)
Design a relational dabase and system architecture	Section 2.1, Section 6
Implement viewing, filtering and mapping of places	FR-01–FR-05
Implement a one-day visit-plan builder	FR-06–FR-08
Implement administrator add/edit/delete functionality	FR-10–FR-14

8. Appendices

Appendix A: Glossary

Term	Definition
Place of interest	A location (religious, nature, heritage or cultural site) documented in the system for tourists to view and plan visits to.
Visit plan	A tourist's selected, ordered set of places forming a single day's itinerary within the system.
Administrator	An authenticated user role responsible for maintaining place data.

Appendix B: Open Issues

B.1 Use Case Diagram

Figure 1 shows the two actors of the system Tourist and Administrator and the use cases each can perform. "View Place Details" includes "View Location on Map" since a place's map location is always shown as part of its detail view. All Administrator place-management use cases include "Admin Login", since authentication is required before any management action.

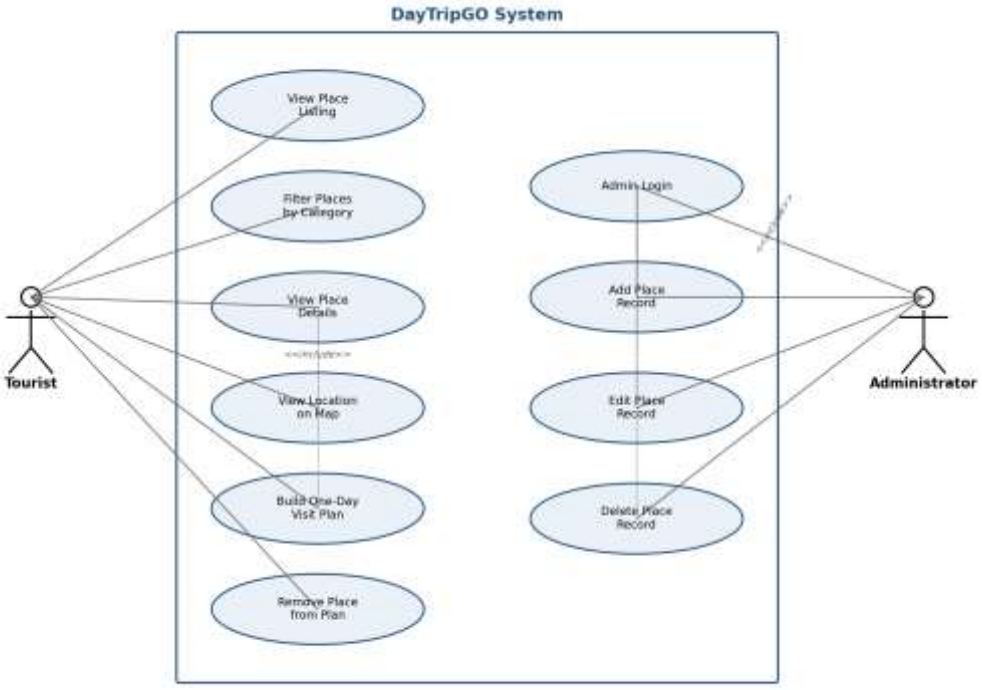


Figure 1: Use Case Diagram for DayTripGO

B.3 Class Diagram

Figure 2 shows the main classes in the system, their key attributes and methods, and the relationships between them, consistent with the entity-relationship diagram in Section 6.

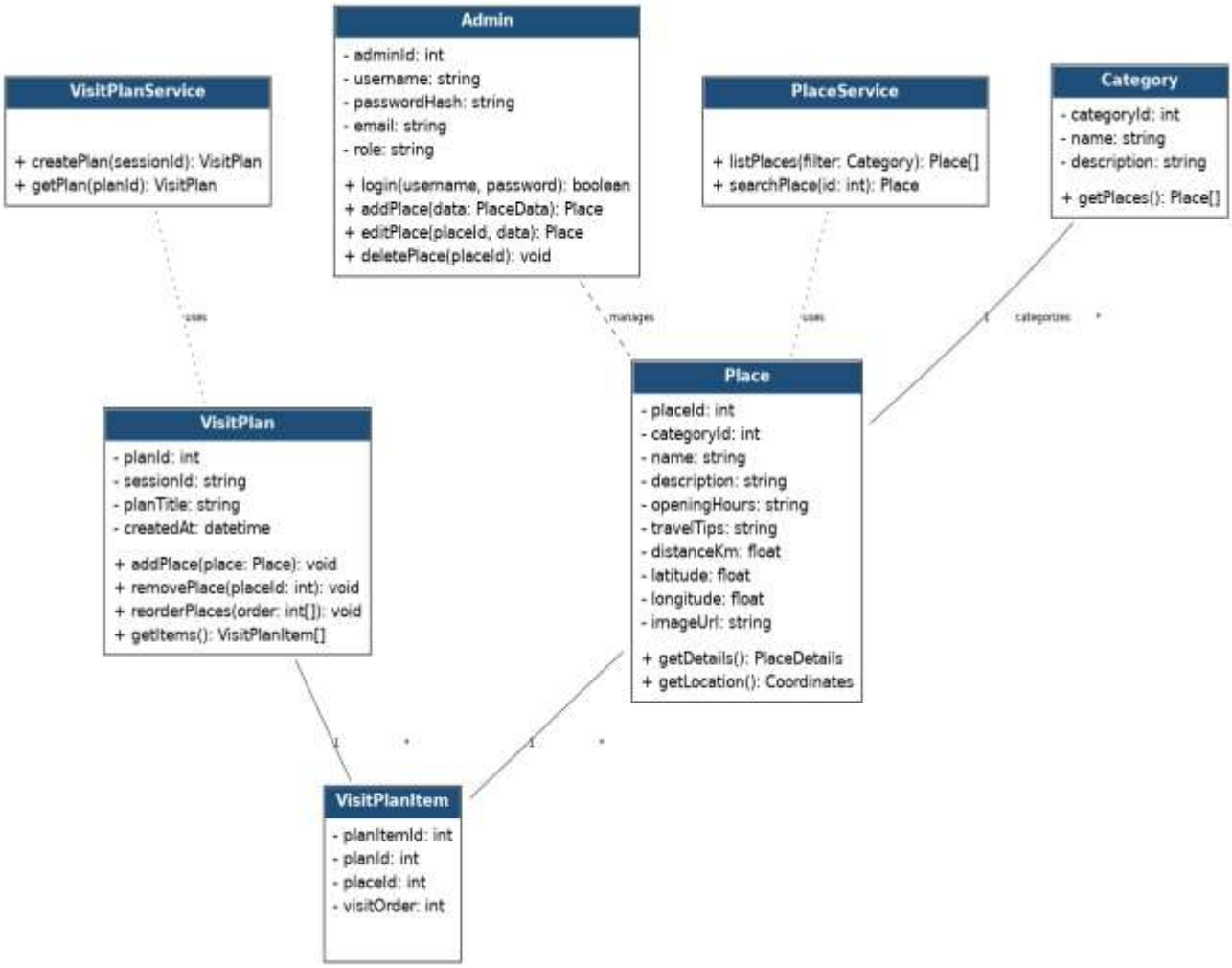


Figure 2: Class Diagram for DayTripGO

B.3 Sequence Diagram

Figure 3 illustrates the interaction between the Tourist, the frontend UI, the backend services and the database for the "Build One-Day Visit Plan" use case (UC-05), from selecting a place through to the visit plan being updated.



Figure 3: Sequence Diagram — Build a One-Day Visit Plan

B.4 Entity-Relationship Diagram

The system uses a relational database (PostgreSQL, hosted on Supabase) with five core entities: Category, Place, VisitPlan, VisitPlanItem and Admin. Category has a one-to-many relationship with Place. VisitPlan and Place have a many-to-many relationship, resolved through the VisitPlanItem join entity, which also stores the visit order for the reordering feature (FR-13). Admin has a one-to-many relationship with Place representing which administrator created or last managed a given record.

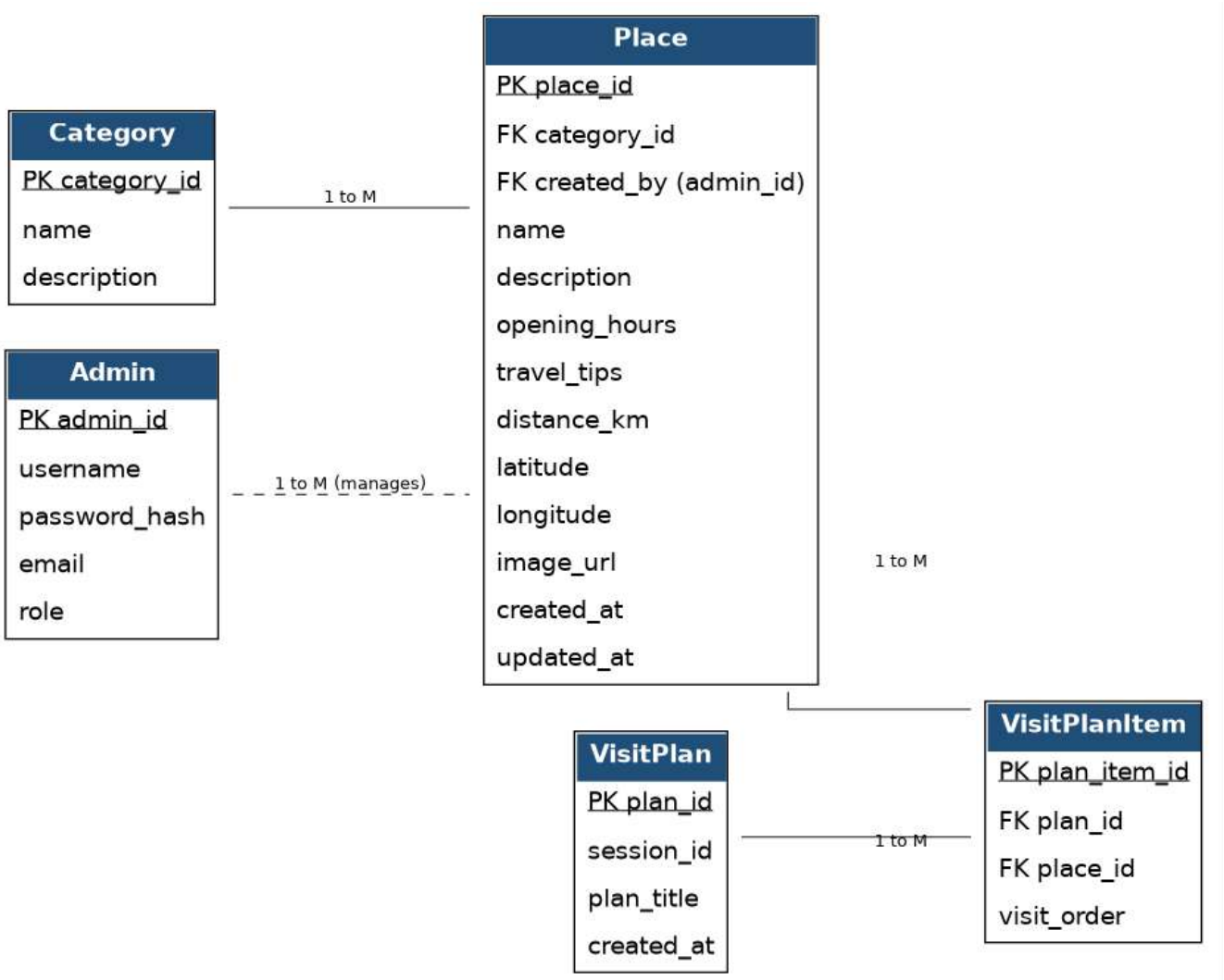


Figure 4: Entity-Relationship Diagram for DayTripGO

Entity Descriptions

Entity	Purpose
Category	Stores the fixed set of place categories (religious, nature, heritage, cultural) used for filtering.
Place	Stores each place of interest's descriptive, locational and operational details.
VisitPlan	Represents a single tourist's one-day visit plan, identified by a browser session rather than a user account.
VisitPlanItem	Join entity linking a VisitPlan to the Places selected for it, including their visit order.
Admin	Stores administrator login credentials and identifies who manages each Place record.

Appendix C: Open Issues

- Final confirmation of the technology stack (Node.js/Express + MySQL vs. PHP alternative) pending module approval.
- Exact distance verification for the Kosgoda Sea Turtle Conservation Project, which sits close to the 25 km boundary.
- Selection of the specific map-embedding API (Google Maps vs. OpenStreetMap/Leaflet) pending cost/quota review for a student project.